

Derbyshire Shared Care Pathology Guidelines

Use of Cardiac Markers in Primary Care

Purpose of Guideline

The management of patients in primary care with suspected myocardial infarction

Definition

Myocardial infarction is defined if the following criteria are satisfied:

Rise and fall of Troponin T and at least one of the following:

- 1) Typical ischaemic chest pain
- 2) Pathological Q waves on ECG
- 3) ST elevation or depression
- 4) Coronary occlusion on angiography

The expert consensus definition of Myocardial Infarction (see Ref 1) defines MI as follows:

- Type 1 MI. Caused by an acute atherothrombotic coronary event following plaque rupture
- Type 2 MI. An entity where a condition other than coronary artery disease (CAD) contributes to a critical imbalance between oxygen supply (such as hypoxemia, anaemia, or hypotension) and demand (such as tachycardia, tachyarrhythmias, or hypertension)

Cardiac chest pain

Assessment of patients with acute chest pain in primary care is not recommended

Patients presenting with a suspected acute coronary syndrome with cardiac chest pain of > 15 minutes duration require urgent assessment in hospital

Dial 999 if they have current pain

When is assessment of patients with chest pain appropriate in primary care?

- If patient presents some days after a possible myocardial infarction and no longer has chest pain, a falling Troponin T level may allow a retrospective diagnosis of myocardial infarction (Troponin T can remain elevated for 2 weeks after a large MI). Such patients should be referred to cardiology outpatients. If the Troponin T is over 100 ng/L or there are new ECG changes suggestive of an MI, consideration should be given to discussion with the on-call cardiologist or cardiology registrar.
- Patient refuses to go to hospital for initial assessment but would do so if ECG and/or cardiac markers are found to be abnormal

Cardiac Markers

Troponin T – high sensitive (TnT-HS)

A highly specific marker for detecting myocardial injury from many causes (see below). Following a myocardial infarction, levels rise rapidly and are detectable after 3 - 6 hours, peaking at 18 - 24 hours. They may stay elevated for 2 - 3 weeks.

How to request cardiac markers

- Requests should be marked urgent (for RDH requests send in a red bag)
- Contact details for telephoning abnormal results must be included on the request form

Interpretation of Troponin T-HS

- Acute myocardial damage highly **unlikely** if Troponin-T HS **<14 ng/L** 12 hours after the onset of chest pain. In low-risk patients without ECG changes or ongoing pain, a value of <14ng/L 6 hours after onset of pain indicates acute myocardial damage is unlikely.
- Troponin T-HS **≥14 ng/L** indicates the **increased probability** of myocardial damage



The laboratory will telephone all **Troponin-T HS ≥14 ng/L** Monday to Friday when GP practices are open. The result will usually be telephoned to out of hours GP service when the GP practice is closed.

- Troponin T-HS **≥ 100 ng/L** is usually indicative of acute myocardial damage ***in an appropriate clinical context***
- For Troponin T-HS values **between 14 – 99 ng/L**, **OR** where there is doubt over the diagnosis, it is recommended to re-measure the Troponin-T HS to assess the trend.
 - Many patients with a peak Troponin T-HS level of less than 100 ng/L will have a final diagnosis other than acute coronary syndrome
 - Troponin T-HS can be chronically elevated in patients with renal failure or heart failure. An acute rise can be seen in many illnesses other than acute coronary syndrome such as PE, sepsis, myocarditis, arrhythmia.

Suspected angina

A referral can be made to the rapid access chest pain clinic unit if the history is suggestive of recent onset stable angina.

NOTES

Troponin T-HS will not be added to a request without prior discussion with the requesting doctor or appropriate colleague
Troponin T-HS can be requested on the Anglia ICE system

Contacts

Derby

Duty Biochemist	01332 789383 (8am to 7pm, Mon – Fri)
On Call Consultant Biochemist	Via RDH switchboard, 01332 340131 (24/7)
Cardiology Advice	Via RDH switchboard, 01332 340131
Chest pain assessment unit	Via RDH switchboard, 01332 340131
A&E, MAU and CDC	Via RDH switchboard, 01332 340131(24/7)

OR

ED Nurse in Charge	07799 337674
MAU Nurse in Charge	07917 650751

Chesterfield

Duty Biochemist	01246 512212 (8am to 5pm, Mon – Fri)
On Call Consultant Biochemist	Via CRH switchboard (24/7)
Cardiology Advice	Via CRH switchboard
	Or
	Cardiology CAO, 01246 512407
A&E, EMU and CDU	Via CRH switchboard

Burton

ED	Via QHB Switchboard
Cardiology Advice	
Chest pain assessment unit	
Duty Biochemist	01283 593313 (8.30 am to 4pm, Mon – Fri)
On Call Consultant Biochemist	Via QHB switchboard, 01283 566333

Reference

1. Thygesen K, Alpert JS, Jaffe AS, et al. Third universal definition of myocardial infarction. Glob Heart 2012; 7:275-95. 10.1016/j.gheart.2012.08.001
2. Recent-onset chest pain of suspected cardiac origin: assessment and diagnosis Clinical guideline Reference number:CG95 Published: 24 March 2010 Last updated: 30 November 2016

Document controls

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	V1 April 2012	Dr Julia Baron, Ms Julia Forsyth	New Document
	V2 May 2014	Dr J Baron, Ms J Forsyth, Dr P Blackwell, Mrs H Seddon	Routine Review
	V3 June 2016	Dr J Baron, Ms J Forsyth, Dr P Blackwell, Mrs H Seddon	Routine Review
	V4 June 2018	Dr J Baron, Dr P Blackwell, Mrs H Seddon	Routine Review
	V5 Dec 2020	Dr J Baron, Dr T Azeem, Dr J Cooke, Dr P Blackwell, Mrs H Seddon	Routine Review
	V6 May 2023	Dr P Blackwell, Mrs H Seddon	Routine Review
Approval	Date	Name(s)	Designation
Secondary Care UHDB	14.05.25	Dr J Baron Dr N Ahmed	Consultant Cardiologists
Primary Care	13.05.25	Dr P Blackwell	GP
Laboratory (Derbyshire Pathology)	20.05.25	Ms S Knowles	Consultant Clinical Scientist